

TUPELO, MS, USA

WMO: 723320

Lat: 34.262N

Lon: 88.771W

Elev: 361

StdP: 14.51

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93862

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.1	23.4	6.6	7.8	25.6	10.7	9.6	28.2	21.7	50.8	19.8	47.4	6.7	10	0.415

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	96.0	75.8	93.7	75.7	91.8	75.3	79.3	89.9	78.3	88.7	77.5	87.7	6.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.5	139.8	84.2	75.4	135.0	83.4	74.5	130.8	82.4	43.0	89.8	42.0	89.2	41.0	87.8	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	16.8	15.0	12.6	98.8	4.5	3.8	9.4	101.6	6.7	103.8	4.2	106.0	0.9	108.7
			11.2	81.3	4.5	1.2	8.0	82.1	5.3	82.8	2.8	83.5	-0.4	84.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.8	42.3	46.5	54.2	62.7	70.9	78.3	81.1	80.7	74.7	63.7	52.4	45.3
	DBStd	15.86	10.99	10.09	9.52	7.72	6.42	4.43	3.66	4.15	6.18	8.27	8.88	9.74
	HDD50	826	291	170	66	6	0	0	0	0	0	4	76	213
	HDD65	2865	704	520	351	132	25	0	0	7	127	386	613	
	CDD50	5508	53	72	197	387	649	848	964	953	742	427	149	67
	CDD65	2072	1	3	16	62	209	398	499	488	299	86	9	2
	CDH74	20422	1	12	134	531	1847	3968	5229	5039	2798	800	59	4
	CDH80	8792	0	1	15	89	626	1781	2443	2419	1178	232	8	0
(14) Wind	WSAvg	6.3	7.4	7.7	7.8	7.4	6.1	5.2	4.6	4.7	5.4	5.6	6.3	7.0
(15) Precipitation	PrecAvg	57.2	5.0	5.0	5.5	5.5	5.2	4.8	4.2	3.9	3.7	3.7	4.4	6.4
	PrecMax	86.4	10.7	11.2	9.4	14.5	17.1	11.1	9.7	8.2	10.7	9.9	9.7	14.9
	PrecMin	36.9	1.8	2.4	1.6	2.1	0.6	0.7	1.2	0.4	0.4	0.0	1.3	2.0
	PrecStd	10.1	2.3	2.2	2.0	2.7	3.8	2.8	2.1	2.2	2.7	2.6	2.0	3.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.2	75.9	82.2	85.5	91.4	96.5	98.1	100.7	95.8	89.3	80.2	73.1
		MCWB	62.6	64.2	65.4	68.3	72.9	74.6	76.0	75.5	72.9	69.8	65.6	65.7
	2%	DB	67.6	71.3	77.8	82.7	88.6	93.5	95.0	96.6	92.7	85.6	74.5	68.5
		MCWB	61.0	61.8	63.5	66.9	71.7	75.0	76.9	76.4	73.1	69.5	63.3	62.5
	5%	DB	63.9	67.1	74.0	79.9	86.4	91.4	93.0	93.7	90.1	82.4	71.0	65.1
		MCWB	59.1	59.0	61.0	65.8	70.7	74.7	76.8	75.9	72.9	67.4	62.2	60.3
	10%	DB	60.2	63.2	70.2	77.0	83.8	89.3	91.0	91.1	87.2	78.7	67.6	61.4
		MCWB	55.0	55.8	59.3	64.3	69.9	74.1	76.4	75.6	72.3	66.1	59.8	57.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.5	66.7	68.7	72.5	76.6	79.3	80.5	80.9	77.7	75.2	69.5	67.6
		MCDB	69.0	72.6	76.5	81.4	84.5	89.8	91.2	93.1	88.2	81.8	74.1	72.3
	2%	WB	62.7	64.2	65.8	69.9	74.6	78.0	79.5	79.3	76.3	73.0	67.0	64.4
		MCDB	66.4	69.3	74.4	78.3	84.1	88.6	90.1	90.1	86.3	81.2	72.1	67.6
	5%	WB	59.8	61.0	63.7	68.4	73.2	77.0	78.5	78.1	75.2	70.9	64.3	61.4
		MCDB	63.5	65.8	71.3	76.5	82.6	87.3	88.8	88.6	84.6	78.3	69.2	64.5
	10%	WB	56.0	57.7	61.4	66.5	71.7	75.9	77.7	77.2	74.0	68.8	61.5	57.5
		MCDB	59.2	61.8	68.4	74.2	80.6	85.4	87.8	87.2	82.6	75.3	66.6	60.9
(35) Mean Daily Temperature Range	5% DB	MDBR	18.5	19.6	21.3	21.7	20.4	19.2	18.4	19.3	21.1	22.6	21.2	18.0
		MCDBR	20.7	23.6	25.0	23.6	22.8	22.0	20.7	22.6	23.3	25.9	24.5	20.5
		MCWBR	15.3	15.6	13.7	11.1	9.1	7.4	6.7	6.7	7.8	11.4	15.0	15.1
	5% WB	MCDBR	18.1	20.1	21.0	18.8	19.2	18.7	18.5	18.8	19.0	19.3	19.3	17.4
		MCWBR	15.0	15.1	13.4	10.6	8.8	7.7	7.1	6.9	7.7	10.7	14.1	14.4
(40) Clear-Sky Solar Irradiance	taub		0.316	0.323	0.360	0.393	0.426	0.459	0.503	0.506	0.431	0.368	0.338	0.316
	taud		2.515	2.491	2.414	2.320	2.271	2.195	2.105	2.088	2.289	2.440	2.467	2.526
	Ebn at Noon		285	294	290	284	274	264	251	248	263	274	273	277
	Edh at Noon		26	30	35	41	43	46	50	50	39	31	28	25
(44) All-Sky Solar Radiation	RadAvg		797	955	1323	1685	1884	2026	1934	1780	1559	1245	906	683
	RadStd		66	127	120	110	118	139	111	113	131	162	90	69

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (23 neighbors)	+0.67	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+198	+142

Nomenclature: See separate page